

Deep Learning Models of Loan Portfolios and Asset-Backed Securities



Felipe Pombal Fritsch
Reuben College
University of Oxford

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Abstract

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Chapter 1

Introduction

1.1 The modelling object

A mortgage is a small dynamical system. In any month it occupies one of a small number of *states* — it is current on its payments, or 30, 60, or 90-plus days past due, or it has terminated through prepayment or a credit event — and between consecutive months it moves between these states with probabilities that depend on the loan’s characteristics and the economic environment. The cashflows of any mortgage portfolio or mortgage-backed security are completely determined by these movements: a prepayment returns principal early (a risk to the investor when rates have fallen), while the path from delinquency through foreclosure destroys value. Modelling the *conditional transition distribution* is therefore the natural primitive for both credit and prepayment risk.

Formally, let $S_{i,t} \in \mathcal{S}$ denote the state of loan i in month t , and let $x_{i,t} \in \mathbb{R}^d$ collect everything observable about the loan and its environment at t . The object of interest is

$$p_\theta(j \mid x_{i,t}) \approx \mathbb{P}(S_{i,t+1} = j \mid x_{i,t}), \quad j \in \mathcal{S}, \quad (1.1)$$

a multinomial classification of next month’s state. Every model in this dissertation — from a frequency-count transition matrix to an ensemble of deep neural networks — is an estimator of (1.1); they differ only in how flexibly they allow the probabilities to depend on the covariates.

The central empirical claim of Sirignano et al. [5], which this dissertation tests on a different and cleaner dataset, is that the dependence of (1.1) on x is strongly *nonlinear* and features economically meaningful *interactions* between covariates. Linear models misspecify this dependence, and the misspecification compounds when

loan-level probabilities are aggregated to pool-level cashflows — the level at which mortgage-backed securities are priced.

1.2 Design of the study

The analysis is organised as a horse race between estimators of (1.1) of increasing flexibility, in three phases. First, an exploratory phase establishes the dataset’s scope and produces *model-free* evidence of nonlinearity and covariate interactions (Chapter 3). Second, loan-level models are estimated and compared strictly out of sample under a rolling, expanding-window design: an empirical transition matrix, multinomial logistic regression, deep neural networks of varying depth, and an ensemble (Chapters 2 and 4). Third, the fitted monthly models are composed into twelve-month horizon predictions and compared at the pool level — predicted versus realised prepayment and delinquency counts, and their translation into valuation units (Chapter 4).

1.3 Contributions

The dissertation deliberately adopts a replication *design* — the cleanest way to attribute findings to data and regime rather than to modelling idiosyncrasies — but four elements are original.

1. **A prime, conforming testbed.** The evidence in Sirignano et al. [5] comes from private-label data with a large subprime component, 1995–2014. This study asks whether the deep-learning advantage survives in the clean GSE credit box, where events are rarer and relationships plausibly tamer. An early answer is already in hand: the architecture search at full scale selects *three* hidden layers over the five preferred in the paper — the nonlinearity premium is real but structurally simpler in prime credit.
2. **Eleven test years the literature has not examined**, 2015–2025, including the COVID-era forbearance episode, the 2020–21 refinancing wave, and the 2022–23 rate shock — regimes that post-date the paper’s sample entirely.
3. **A stricter evaluation protocol whose output is itself a finding.** Replacing the paper’s single temporal split with an eleven-window rolling backtest produces a *time series of the value of nonlinearity*: in which market regimes does the deep model’s edge widen, compress, or invert? The single-split design cannot ask this question.

4. **An economic translation for the pool level.** Predicted and realised pool prepayments are converted into each pool’s prepayment-speed path, run through a pass-through cashflow engine, and compared as errors in cashflow timing and price — restating the model comparison in the units an investor in mortgage pools actually uses, rather than in log-likelihood alone.

1.4 Outline

Chapter 2 develops the methodology: the transition model and its estimation at scale, the rolling out-of-sample design, the model hierarchy, the evaluation criteria, and the pool-level framework including the economic translation. Chapter 3 describes the dataset, the construction of the estimation panel, the macroeconomic covariates, and the exploratory evidence that motivates nonlinear modelling. Chapter 4 reports the results. Chapter 5 concludes.

Chapter 2

Methodology

2.1 The transition model

The supervised object is the one-month transition distribution (1.1) over the seven states defined in Section 3.4. In this dataset the terminal states (*Prepaid*, *Foreclosure*, *REO*) are absorbing — they are observed only in a loan’s final month — so the origin set comprises the four live states and the transition matrix is 4×7 rather than 7×7 (a documented difference from 5, harmless for the model comparison since every competing model conditions on the same origin set). A single multinomial model is fitted with the current state as an input feature, rather than four separate per-origin models. Some transitions are structurally impossible — delinquency deepens at most one bucket per month, so *Current* cannot reach 60 DPD directly — and rather than hard-coding a mask, the models are left to learn these near-zeros; the evaluation suite asserts that the predicted mass on impossible cells is negligible, a useful internal consistency check.

2.2 Estimation at scale: sharding and minibatch randomisation

Stochastic gradient descent assumes minibatches are approximately random draws from the training distribution. The panel, however, is stored grouped by acquisition vintage: read sequentially, it yields batches composed entirely of, say, 2007-vintage loans, whose risk profile is wildly atypical, and the resulting gradient bias changes within an epoch. The textbook remedy — globally shuffling the dataset each epoch — is infeasible for $\sim 10^9$ rows on external storage.

The pipeline instead assigns every row a *shard*,

$$\text{shard}(i) = h(\text{LoanID}_i) \bmod N, \quad N = 256, \quad (2.1)$$

where h is a fixed-seed 64-bit hash. Three properties follow, each load-bearing. First, a good hash behaves like an independent uniform draw per loan identifier, so each shard is an unbiased $\frac{1}{N}$ sample of *loans*, mixing all vintages and geographies (verified empirically: rows and exact distinct loans per shard are uniform to coefficients of variation of 0.26% and 0.21%, and per-shard vintage composition matches the panel marginal). Second, because the hash is keyed on the loan — not the row — every monthly observation of a loan lands in the same shard, so shard-based subsets never split a loan between training and evaluation. Third, the assignment is a pure function of the loan identifier: rebuilding the panel reproduces every shard exactly, with no random state to persist.

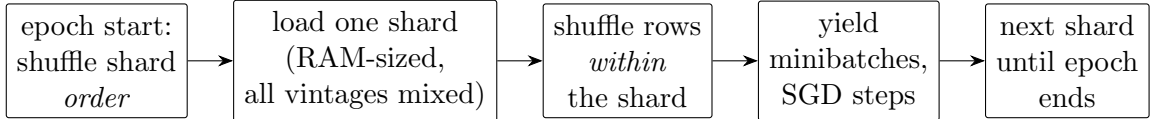


Figure 2.1: *The two-level shuffle*. Random shard order across the epoch, full shuffle within each RAM-sized shard. Every minibatch mixes loans from all vintages at streaming cost; a global shuffle of 10^9 rows is never performed.

Training then follows the two-level shuffle of Figure 2.1. It is not exactly a uniform draw — rows in a batch share a shard — but because each shard is already an unbiased cross-section of loans, the residual within-shard correlation is negligible for gradient quality.

2.3 Covariates and feature construction

Features comprise the loan-level fields and external macroeconomic covariates described in Sections 3.3 and 3.3.3. Continuous covariates are standardised using moments estimated on each training window only (Section 2.4); heavily skewed balances enter in logarithms. High-cardinality categorical fields (property state, metropolitan area, three-digit zip code) are mapped to learned dense embedding vectors rather than one-hot columns, with categories unseen in training routed to a reserved index. Missingness indicators accompany sentinel-scrubbed fields, since the fact of missingness is itself predictive in credit data.

Two engineered covariates carry most of the economic content:

$$\text{(refinancing incentive)} \quad \text{incentive}_{i,t} = r_{i,t}^{\text{cur}} - m(t), \quad (2.2)$$

$$\text{(mark-to-market LTV)} \quad \text{ltv}_{i,t}^{\text{mtm}} = \frac{\text{UPB}_{i,t}}{\text{UPB}_{i,0}} \cdot \text{LTV}_{i,0} \cdot \frac{H_{s(i)}(t_0(i))}{H_{s(i)}(t)}, \quad (2.3)$$

where $m(t)$ is the prevailing 30-year mortgage rate, UPB denotes the unpaid principal balance (so $\text{UPB}_t/\text{UPB}_0$ is the fraction of the loan still owed), and $H_s(\cdot)$ is the state- s house-price index at the indicated month. Equation (2.2) measures how far the borrower’s refinancing option is in the money — the dominant prepayment driver [4] — and (2.3) approximates the borrower’s current leverage, the key default driver [1]. Raw calendar time is deliberately excluded as a feature: test years are by construction unseen, so a model leaning on “year” would extrapolate arbitrarily; all time variation must flow through observable, recurring channels (the macro block, seasonality encoded as month-of-year, and loan age).

2.4 Rolling out-of-sample design

Credit models are used prospectively, so the evaluation is temporal: train on the past, predict the future. One subtlety deserves emphasis because it is a one-character bug with real leakage consequences: the label of the example built at feature month t is the state at $t+1$, so “training on data up to month T ” must mask on the *label* month — include the example only if $t < T$ strictly, so that its label month $t+1 \leq T$.

The primary design is a rolling, expanding-window backtest with one window per test year $k = 2015, \dots, 2025$:

$$\underbrace{\text{train: labels } \leq \text{Dec}(k-2)}_{\text{expanding, starts 2000}} \quad \underbrace{\text{validation: labels in year } k-1}_{\text{early stopping only}} \quad \underbrace{\text{test: labels in year } k}_{\text{never touched in fitting}}$$

Three rules keep this honest and affordable. First, *tune once, freeze, roll*: architecture and regularisation hyperparameters are selected once, on the $k=2015$ tuning window, then frozen across all windows; per-window validation years serve only early stopping. Re-tuning every window would multiply compute by the grid size and constitute a subtle multiple-testing problem. Second, *window-local preprocessing*: feature scalers, embedding vocabularies, and any statistic estimated from data are re-fitted within each window’s training slice; standardising with full-sample moments would leak future means and variances into the past. Third, *pooled and per-year reporting*: each test row in 2015–2025 is predicted exactly once, by a model that never saw it, and

metrics are reported per test year and pooled over all years. Regime shifts — COVID in 2020–21, the rate shock of 2022–23 — thus appear as legible test-year effects rather than hinging on a single arbitrary cutoff.

Class thinning with importance weights. *Current-to-Current* transitions are 96.3% of loan-months (Table 3.4) and individually nearly uninformative. The training export therefore retains every row whose origin or destination is not *Current*, but keeps *Current*→*Current* rows only with probability p , attaching weight $w_i = 1/p$ to kept rows ($w_i = 1$ otherwise). The training objective is the weighted negative log-likelihood

$$\hat{\mathcal{L}}(\theta) = \frac{1}{\sum_i w_i} \sum_{i \in \text{kept}} w_i [-\log p_\theta(y_i | x_i)], \quad (2.4)$$

a Horvitz–Thompson estimator of the full-sample loss [3]: each row is kept with known probability and weighted by its inverse, so the fitted model still estimates the *true* conditional probabilities — unlike naive downsampling, which would silently inflate every rare-event probability. Validation and test slices are never thinned, and a QA step compares ensemble-average predicted base rates against unthinned panel frequencies. Evaluation uses a fixed, loan-disjoint shard block ($\sim 20\%$ of loans, the same loans in every window and for every model), so that likelihood comparisons are exact rather than sampled apart.

2.5 The model hierarchy

The benchmarks are ordered by how flexibly they let (1.1) depend on covariates; each strictly nests or dominates the previous, which is what makes the comparison interpretable.

2.5.1 Empirical transition matrix

The simplest estimator ignores covariates entirely: conditional only on the origin state u , the destination distribution is estimated by smoothed relative frequencies in the window’s training slice,

$$\hat{P}_{uv} = \frac{N_{uv} + \alpha}{N_{u\cdot} + \alpha |\mathcal{S}|}, \quad \alpha = \frac{1}{2}, \quad (2.5)$$

where N_{uv} counts $u \rightarrow v$ transitions. The Laplace term α guarantees no test transition receives probability zero (a single unsmoothed empty cell would make the test

log-likelihood infinite). This time-homogeneous Markov chain is the floor of the comparison. A “bucketed” variant conditioning additionally on credit-score and incentive terciles shows how far cheap, nonparametric conditioning goes before any model fitting.

2.5.2 Multinomial logistic regression as a zero-layer network

The workhorse of the empirical mortgage literature posits log-linear odds:

$$p_{\theta}(j \mid x) = \text{softmax}_j(Wx + b) = \frac{\exp(w_j^{\top}x + b_j)}{\sum_{j'} \exp(w_{j'}^{\top}x + b_{j'})}. \quad (2.6)$$

Crucially, (2.6) is exactly a neural network with zero hidden layers, and it is implemented as such — same data loader, same loss (2.4), same evaluation code as the deep models, with a one-time numerical cross-check against a standard library implementation. This is an experimental-control choice: when the deep network beats the logit, the difference is attributable to architecture alone, not to differences in pre-processing, optimisation or data. An “augmented” logit with hand-crafted nonlinear terms (squared loan age, splined incentive) quantifies how much of the gap manual feature engineering can close.

2.5.3 Deep neural networks

The deep model composes L hidden layers,

$$z^{(0)} = \phi(x), \quad z^{(\ell)} = \sigma(W^{(\ell)}z^{(\ell-1)} + b^{(\ell)}), \quad p_{\theta}(\cdot \mid x) = \text{softmax}(W^{(L+1)}z^{(L)} + b^{(L+1)}), \quad (2.7)$$

with rectified-linear activations $\sigma(u) = \max(u, 0)$ and the input map ϕ of Section 2.3. Depth is searched over $L \in \{1, 3, 5, 7\}$ anchored at the five-layer geometry preferred by Sirignano et al. [5] (200 units, then 140 per layer); layer widths are inherited from that paper rather than searched — depth and regularisation are the first-order capacity knobs — with a half/paper/double width sweep at the selected configuration as a completeness check (Appendix A). Each hidden unit is a learned “soft feature”; depth lets the network build feature hierarchies, which is where interaction effects such as incentive×credit-score live — the model discovers them rather than being told.

Three regularisation devices mirror the paper: *dropout* [6] on hidden layers (rates $\{0, 0.2, 0.5\}$), an ℓ_2 penalty ($\lambda \in \{0, 10^{-5}, 10^{-4}\}$), and *early stopping* on the window’s validation likelihood — the only use the validation year is put to. Finally, the headline

model is an *ensemble*: $K=8$ independently initialised networks trained on resampled shard subsets, with predicted probabilities averaged, $\bar{p}(j \mid x) = \frac{1}{K} \sum_k p_{\theta_k}(j \mid x)$. Averaging reduces the variance component of error from stochastic training and finite data; the marginal value of each additional member is traced in an ensemble-size curve. Because eight fits per window is the dominant compute cost, the ensemble runs on five regime-spanning key windows ($k \in \{2015, 2019, 2020, 2023, 2025\}$), with the single best network everywhere.

2.6 Evaluation

The headline criterion is the out-of-sample mean negative log-likelihood,

$$\text{NLL} = -\frac{1}{N_{\text{test}}} \sum_{i \in \text{test}} \log p_{\theta}(y_i \mid x_i), \quad (2.8)$$

on identical, frozen, unthinned test rows for every model. It is the right primary metric for two reasons: it evaluates the full predicted distribution over seven destinations rather than a binarised event, and it is a strictly proper scoring rule — uniquely minimised in expectation by the true conditional probabilities, so a model cannot improve it by distorting its probabilities. Two complementary lenses guard against failure modes a single number hides: *discrimination*, measured for each origin–destination pair by the one-versus-rest area under the ROC curve (the probability that a randomly chosen loan-month that truly makes the transition receives a higher predicted probability than one that does not); and *calibration*, measured by reliability diagrams comparing predicted probabilities to realised frequencies within prediction buckets. A model can rank well yet be systematically over- or under-confident; calibration is what the pool-level counting of Section 2.7 actually consumes.

2.7 From loans to pools

A mortgage pool’s cashflow is the sum of its loans’ events; a mortgage-backed security (MBS) is priced off exactly such sums. Sirignano et al. [5] argue the nonlinearity advantage *compounds* rather than washes out at the pool level, because linear misspecification correlates across loans sharing characteristics.

2.7.1 Multi-month prediction by matrix composition

The monthly model is rolled forward by composing transition matrices. For loan i alive at anchor month t_0 with state distribution ρ_0 (a one-hot vector), the model

supplies for each future month h a 7×7 matrix $P_h^{(i)}$ whose four non-terminal rows are the model’s predictions at the deterministically evolved covariates and whose terminal rows are the identity (absorbing states). Then

$$\rho_h = \rho_0 \prod_{h'=1}^h P_{h'}^{(i)}, \quad \mathbb{P}(\text{loan } i \text{ prepaid by } t_0+H) = [\rho_H]_{\text{Prepaid}}, \quad (2.9)$$

with horizon $H = 12$ months. Because terminal states are absorbing and the full state distribution is tracked, (2.9) is exact under the model — no Monte Carlo simulation is required. Covariates evolve deterministically (age and remaining term advance, seasonality follows the calendar) while balances and the macroeconomic block are frozen at t_0 : predictions are explicitly *conditional on the anchor-date environment*. For “ever reaches 60+ days past due within the horizon” a first-passage construction is used: delinquency states can be entered and cured out of, so month- H occupancy undercounts loans that visited and recovered; making the target set absorbing in a modified chain yields the probability of ever hitting it.

2.7.2 Pools and pool-level prediction

Pools are formed two ways at each window anchor: characteristic buckets (credit score $\times$ original rate $\times$ loan-to-value cells, the criteria a securitisation structurer would use) and random pools (B pools of N loans), which give a bucketing-free comparison. With per-loan horizon probabilities q_i from (2.9), the predicted event count in pool $\mathcal{P}$ and its approximate distribution are

$$\hat{C}_{\mathcal{P}} = \sum_{i \in \mathcal{P}} q_i, \quad C_{\mathcal{P}} \sim \mathcal{N}\left(\sum_i q_i, \sum_i q_i(1 - q_i)\right), \quad (2.10)$$

the Gaussian being the standard approximation to the Poisson–binomial distribution of a sum of independent non-identical Bernoulli variables (conditional independence given covariates is the maintained assumption, inherited from the paper). Models are compared on predicted versus realised counts, with interval coverage testing the distributional claim in (2.10), not just its mean.

2.7.3 Economic translation: prepayment speed, cashflow timing, and price

The final step converts counts into valuation units; each term is defined as it first appears. Averaging the loans’ monthly prepayment probabilities with each loan weighted by its unpaid principal balance gives the pool’s **prepayment-speed path**:

the model’s month-by-month forecast of what fraction of the pool’s remaining principal prepays — the units and weighting that pool cashflows actually obey, since a large loan returns proportionally more principal than a small one. (Market shorthand, used hereafter: the monthly fraction is the *single monthly mortality*, SMM; annualised it is the *conditional prepayment rate*, CPR.) The realised path is computed identically from the panel. A deterministic cashflow engine turns any speed path into pool cashflows, treating the pool as a *pass-through* (investors receive the pool’s interest and principal as it is paid): scheduled amortisation plus forecast prepayments, month by month, with the final month’s speed held constant beyond the horizon. From the cashflows follow two valuation summaries: **weighted-average life** — the average time a dollar of principal remains outstanding, i.e. the timing of money returned — and **price**, the discounted value of the cashflows, with every model’s cashflows discounted at the same fixed rate so that any price difference comes from the prepayment forecasts alone. The comparison metrics are then a speed error (percentage points of CPR), a timing error (months of weighted-average life), and a price error (per 100 of face value), per pool, model, and anchor. The engine is unit-tested against two closed forms (with no prepayment a pool is a textbook annuity; a constant speed has a closed-form survival schedule). With the macroeconomic environment frozen at the anchor and a common fixed discount rate, the exercise isolates the *prepayment-model component* of valuation error — deliberately simpler than the market’s full option-adjusted-spread machinery.

Why pricing rather than portfolio sorts. Sirignano et al. [5] demonstrate economic significance with a portfolio-sort exercise: rank pools by predicted prepayment risk, form best- and worst-decile portfolios, and compare their realised outcomes. This study runs the pricing exercise instead, and the choice deserves a plain explanation. The two designs test different properties of the same predictions. A sort uses only the *ordering* of pools — it is blind to probability levels, so a model whose hazards are uniformly twice the truth sorts exactly as well as a perfectly calibrated one. Pricing uses the *levels*: weighted-average life and price depend on the actual magnitude of the predicted prepayment path. Ranking-only evidence is the natural choice where levels are inherently untrustworthy (the classic asset-pricing setting); here, by contrast, the transition probabilities are explicitly calibration-checked (Section 2.6), so a ranking-only exercise would discard the model’s strongest property — and the practical consumer of a pool-level mortgage model (pricing a pool, marking it to market, hedging a position in it) consumes levels, not ranks. Running both would

add little: each is a deterministic re-presentation of the same per-loan probabilities, their respective properties are already tested more rigorously upstream (discrimination by the AUC tables, calibration by the reliability diagrams), and the bucket-level price-error map conveys what a sort would dramatise — *where* the linear model goes wrong — while also conveying by *how much*, in the units the market trades. The sort exercise is therefore noted as future work rather than duplicated.

2.8 Reproducibility

Every model fit writes a self-describing run record (configuration, code version, data-manifest hash, window identity, metrics), so any number reported in Chapter 4 can be traced to the exact code and data slice that produced it. Correctness-critical components carry unit tests: the leakage mask, the importance weights, the mark-to-market LTV construction, the roll-forward engine against a closed-form chain, and the cashflow engine against the annuity closed forms. Appendix B records the computing environment and data snapshot conventions.

Chapter 3

Data

3.1 Data Source

This study employs the Fannie Mae Single-Family Loan Performance (SFLP) dataset, a publicly available research dataset maintained by the Federal National Mortgage Association (Fannie Mae) and updated on a quarterly basis. The dataset was originally released in April 2013 and contains origination and monthly performance records for a subset of single-family conventional mortgage loans acquired by Fannie Mae since January 2000. New loan acquisitions and performance data are added with an approximate four-month lag following the end of each calendar quarter [2].

The data are organised as a *loan-month panel*: each observation corresponds to a single mortgage loan in a single monthly reporting period, beginning at the month of acquisition and continuing until the loan terminates through full prepayment, foreclosure, Real Estate Owned (REO) disposition, or contractual maturity, or until the data cut-off date. This longitudinal structure is well-suited to the competing-risks framing of Sirignano et al. [5], which the present study adapts to the Fannie Mae data.

Raw files are distributed as pipe-delimited CSV without a header row, organised by acquisition quarter and identified by the convention `YYYYQn.csv`. Column identities are positional, fully specified in the Fannie Mae Single-Family Loan Performance Data File Layout and Glossary. Each row contains 113 fields encoding static origination attributes, dynamic monthly performance measures, and post-termination expense and proceeds data. The estimation sample employed in this study is the *complete* vintage set from a single quarterly release: 104 acquisition-quarter files covering cohorts from 2000Q1 onwards, with monthly reporting periods from January 2000 through December 2025. The assembled panel comprises **3,312,456,883 loan-month observations across 57,562,668 loans** — performance histories spanning

four interest-rate cycles, the 2008–11 foreclosure crisis, the COVID-era forbearance episode, and the 2022–23 rate shock.

3.1.1 From raw files to the estimation panel

Three distinct concepts in this dataset are colloquially called a “quarter,” and distinguishing them prevents the most common assembly errors. The *acquisition vintage* identifies a file: each `YYYYQn.csv` contains the loans Fannie Mae acquired in that quarter, carrying their entire subsequent monthly history, and a loan appears in exactly one file — so concatenating all files assembles the full sample with no duplication. The *monthly reporting period* identifies a row, and this time dimension — not redundancy — is why the dataset is large. The *release* identifies the download: Fannie Mae republishes the entire dataset quarterly with extended histories, so all files must come from one release lest vintages carry inconsistent right-censoring dates; an ingestion audit verifies a uniform cut-off across all 104 files.

Because no single file fits in memory (~ 800 GB of raw CSV in total), a six-stage pipeline converts the files into a compressed, columnar store queried out of core: an integrity audit, streaming conversion, type and sentinel cleaning, construction of the transition panel with the outcome variable of Section 3.4, sampling utilities, and a reconciliation step that verifies row counts against the raw files. Columnar storage means a query touching six of the 113 fields reads roughly five per cent of the bytes, which is what makes the full-panel descriptive statistics and exploratory analysis of Sections 3.5 and 3.6 computationally cheap.

3.2 Population and Eligibility

The primary dataset is not a census of all loans in Fannie Mae’s guarantee book; it is a curated research sample subject to documented eligibility and exclusion criteria [2]. The population is restricted to fixed-rate, fully amortising, full-documentation single-family conventional mortgage loans with original terms of at least five years and less than 35 years, originated on or after 1 January 1999 and acquired by Fannie Mae on or after 1 January 2000.

A number of product types are explicitly excluded. Adjustable-rate mortgages, interest-only loans, balloon-amortisation mortgages, and loans carrying prepayment penalties are absent from the dataset. Also excluded are loans with original loan-to-value (LTV) ratios exceeding 97 per cent, Alt-A and other reduced-documentation products, government-insured loans (FHA and VA), loans subject to long-term standby

commitments or third-party risk-sharing arrangements other than primary mortgage insurance, and loans acquired on a negotiated bulk basis. Loans with non-standard payment due dates and loans that liquidated in the same month as acquisition are similarly excluded. Fannie Mae’s Home Affordable Refinance Program (HARP) loans are likewise absent from the primary dataset; when a loan in the primary dataset was subsequently refinanced through HARP, its termination is recorded as a standard prepayment, and its post-refinance performance is not tracked here. This is discussed further in Section 3.4.

These restrictions yield a sample of conventionally underwritten fixed-rate mortgages broadly representative of the agency conventional loan market under post-crisis underwriting standards. The exclusion of adjustable-rate and interest-only products limits the range of prepayment incentive dynamics captured relative to the full-spectrum dataset used by Sirignano et al. [5]; this is discussed in Section 3.7. Table 3.1 summarises the principal eligibility criteria and exclusions.

Criterion	Requirement / Exclusion
Origination date	On or after 1 January 1999
Acquisition date	On or after 1 January 2000
Product type	Fixed-rate, fully amortising only
Loan term	$5 \leq \text{term} < 35$ years at origination
Documentation	Full documentation only
LTV at origination	$\leq 97\%$
<i>Excluded: product</i>	ARM, interest-only, balloon, prepayment-penalty loans
<i>Excluded: credit class</i>	Alt-A, reduced-documentation, streamlined processing
<i>Excluded: guarantee</i>	Government-insured (FHA, VA) loans
<i>Excluded: structure</i>	Long-term standby commitments; bulk acquisitions; lender-recourse loans (except primary MI)
<i>Excluded: programme</i>	HARP and Refi Plus loans
<i>Excluded: payment</i>	Bi-weekly and non-first-of-month due dates
<i>Excluded: timing</i>	Loans liquidated in month of acquisition

Table 3.1: *Eligibility criteria for the Fannie Mae SFLP primary dataset.* Source: Fannie Mae Single-Family Loan Performance Dataset FAQs [2].

3.3 Variables

The 113 fields in each monthly observation can be organised into three functional groups: static origination attributes, dynamic monthly performance attributes, and

post-termination resolution data. The model uses a subset of these fields as covariates. Two fields — the Current Loan Delinquency Status and the Zero Balance Code — serve as the primary inputs from which the outcome variable is constructed.

3.3.1 Origination Attributes

Static origination attributes are recorded at loan delivery and do not change over the life of the loan absent a modification. The borrower’s FICO credit score at origination is available for both the primary borrower and, where applicable, a co-borrower; prior to the January 2015 dataset release only the lower of the two scores was disclosed, and a missingness indicator is constructed accordingly. Continuous origination covariates include the original debt-to-income (DTI) ratio, original LTV ratio, original combined LTV (CLTV, which incorporates subordinate liens), original unpaid principal balance (UPB), original interest rate, and original loan term in months. Categorical origination attributes include the acquisition channel (retail, broker, or correspondent), loan purpose (purchase, rate-term refinance, or cash-out refinance), property type, occupancy status (primary residence, second home, or investment property), property state, Metropolitan Statistical Area (MSA), and three-digit truncated zip code. The truncation of the zip code reflects borrower privacy considerations; full zip codes are not disclosed in the dataset [2].

3.3.2 Dynamic Performance Attributes

Dynamic performance attributes are updated in each monthly reporting period. The current actual unpaid principal balance reflects the outstanding loan balance at each point in time. Loan age, measured in months from the first payment date, and the remaining months to legal maturity are derived from the contractual amortisation schedule. The current interest rate reflects the applicable note rate, updated when a modification changes the loan terms.

Two dynamic fields jointly determine the outcome variable. The **Current Loan Delinquency Status** reports the number of missed monthly payments as a two-character code: 00 denotes a current loan; 01, 02, and codes 03 and above denote 30-, 60-, and 90-or-more-day delinquencies respectively; and XX denotes an unavailable status. The **Zero Balance Code** is populated exclusively in the terminal monthly observation and classifies the manner of loan termination into nine categories, distinguishing between voluntary prepayments, third-party foreclosure sales, short sales, repurchases, REO dispositions, note sales, and non-credit removals. (The dataset

also carries a 24-month payment-history string; it is not used as a model input here, since the current state and loan-level covariates already summarise the information the transition model conditions on.)

3.3.3 Externally Sourced Covariates

A small, fixed set of external macroeconomic series is joined to the loan-level panel at feature-construction time (the stored panel itself is never rewritten); Table 3.2 summarises the sources. The Freddie Mac Primary Mortgage Market Survey (PMMS) weekly fixed-rate series, aggregated to calendar-month means, supplies the prevailing market mortgage rate from which the *rate incentive* of Equation (2.2) is computed — the refinancing option value, the primary driver of prepayment behaviour in the empirical mortgage literature [4, 5]. Unemployment rates enter at both national (Bureau of Labor Statistics) and state level, the latter because the cross-sectional dispersion in labour-market conditions (Nevada versus Texas in 2009) is precisely the variation a national series cannot supply. House prices enter through the Freddie Mac House Price Index at national and state level, from which the mark-to-market loan-to-value ratio of Equation (2.3) — the principal determinant of the default option value [1] — and the trailing twelve-month house-price change are constructed. The ten-year Treasury yield completes the set.

Each series is joined with a *publication lag* reflecting when the value was actually observable: rates are published in real time (no lag), month- t unemployment is released early in month $t+1$ (one-month lag), and the house-price index roughly two months after the reference month (two-month lag). Final-revision series are used rather than real-time vintages — a stated limitation, standard for historical studies. Where a state-level value is unavailable (territories; series start dates) the national value substitutes, with an indicator flag carried as a model feature.

Variable	Source	Geography	Monthly value	Lag
30-yr mortgage rate	Freddie Mac PMMS	national	month mean	0 m
10-yr Treasury yield	FRED DGS10	national	month mean	0 m
Unemployment rate	BLS (via FRED)	national + state	as published	1 m
House-price index	Freddie Mac FMHPI	national + state	as published	2 m

Table 3.2: *External macroeconomic covariates and join conventions.* A loan-month at calendar month t is joined to the macro value of month $t - \ell$, where ℓ is the publication lag. Seasonally adjusted versions are used throughout.

As a check on the incentive variable, a market-rate proxy derived from the panel itself — the average note rate of loans originated in each calendar month — tracks the PMMS series with correlation 0.991, root-mean-square deviation 0.198 percentage points, and mean absolute deviation 0.147 percentage points, with a mean gap of essentially zero. The PMMS-based incentive is the primary feature; the panel-derived proxy is retained as a robustness alternative, demonstrating that the incentive variable is not an artefact of one data source.

3.4 Outcome Variable

Following Sirignano et al. [5], the outcome variable is a categorical measure of mortgage state in month $t + 1$, conditional on the loan’s state and covariate vector in month t . Seven mutually exclusive states are defined: *Current*, *30 Days Past Due* (30 DPD), *60 Days Past Due* (60 DPD), *90+ Days Past Due* (90+ DPD), *Foreclosure*, *Real Estate Owned* (REO), and *Prepaid*.

State assignment for each loan-month observation follows a deterministic hierarchy. When the Zero Balance Code is present — indicating the terminal observation for the loan — the termination type is mapped to the Foreclosure, REO, or Prepaid absorbing state as shown in Table 3.3. When the Zero Balance Code is absent, the delinquency status field governs: codes 00 through 03 and above map to Current, 30 DPD, 60 DPD, and 90+ DPD respectively, while **XX** is treated as missing. The state in month $t + 1$ is obtained by a window lead over the reporting period dimension, partitioned by loan identifier. Loans whose final observation falls at or before the data cut-off without a recorded termination event receive a censored indicator and are excluded from the labelled target.

Regarding HARP loans: as noted in Section 3.2, any loan in the primary dataset that was refinanced through the HARP programme terminates with a Zero Balance Code of 01 (prepaid) at the moment of refinancing and is therefore treated as a standard prepayment in the outcome variable. This approximation is defensible given that the loan is extinguished from Fannie Mae’s credit exposure in either case, though it may marginally overstate the rate-incentive sensitivity of the prepayment hazard during the period of peak HARP activity (approximately 2009–2013).

One structural feature of this dataset deserves emphasis: because the Foreclosure, REO, and Prepaid states are derived from the Zero Balance Code — populated only in a loan’s final observation — these three states are *absorbing*. There is no ongoing “in foreclosure” monthly status as in the CoreLogic data of Sirignano et al. [5], where

State	Source field	Condition	Economic interpretation
Current	Delinquency Status	= 00	Borrower is contractually current on all payments.
30 DPD	Delinquency Status	= 01	One missed monthly payment.
60 DPD	Delinquency Status	= 02	Two consecutive missed monthly payments.
90+ DPD	Delinquency Status	$\geq$ 03	Three or more missed payments; serious delinquency.
Foreclosure	Zero Balance Code	02, 03, 15, 97, 98	Credit-loss termination: third-party sale, short sale, note sale, or credit-event repurchase.
REO	Zero Balance Code	= 09	Fannie Mae acquires ownership of the property.
Prepaid	Zero Balance Code	01, 06, 16, 96	Full payoff, contractual maturity, repurchase, or non-credit removal (including HARP exits).

Table 3.3: *Mapping of Fannie Mae data fields to the seven-state outcome variable.* The Zero Balance Code is populated only in the terminal monthly observation; the Delinquency Status governs state assignment in all non-terminal periods.

loans in formal foreclosure are observed month by month and occasionally return to performing status. The origin set here therefore comprises the four live states (Current through 90+ DPD) while destinations span all seven: the transition matrix is 4×7 rather than 7×7 (Section 2.1). Within the live states the data are rich in both directions — delinquency deepens at most one bucket per month, but cures can jump multiple buckets, and a meaningful fraction of seriously delinquent loans return to current. The seven-state formulation captures this in a way that a binary default-versus-prepayment classification would obscure, and the empirical monthly transition matrix derived from each training window serves as the floor benchmark for the model comparison (Section 2.5.1).

3.5 Panel Composition and Summary Statistics

The assembled panel comprises 3,312,456,883 loan-month observations across 57,562,668 loans in 104 acquisition-vintage partitions, with reporting months from January 2000 through December 2025. Table 3.4 reports the distribution of loan-months across the seven states. Two features dominate. First, extreme class imbalance: the Current state accounts for 96.28% of all loan-months, while the credit-event terminals occur

at roughly one in ten thousand observations — the empirical fact behind both the importance-weighted thinning of training data (Section 2.4) and the Laplace smoothing of the empirical benchmark (Section 2.5.1). Second, the rare states are nonetheless abundantly populated in absolute terms (REO alone contributes roughly half a million loan-months), which is what permits stable estimation of rare-transition probabilities at this scale.

State	Share	State	Share
Current	96.28 %	60 DPD	0.31 %
Prepaid	1.25 %	REO	0.014 %
30 DPD	1.15 %	Foreclosure	0.0067 %
90+ DPD	0.98 %		

Table 3.4: *Distribution of loan-months across the seven states*, full panel, 2000–2025 (3.31 billion loan-months).

Pooled over time, the monthly transition rates out of Current are 1.27% to Prepaid and 0.64% to 30 DPD. The pooling, however, hides the economics: Figure 3.1 plots the monthly series over 2000–2025, displaying the 2003 and 2020–21 refinancing waves in the prepayment rate, the 2008–11 default wave in the delinquency rates, and the sharp 2022–23 collapse in prepayments as market rates rose. These regime shifts — rather than any single structural break — are the empirical justification for the rolling backtest design of Section 2.4. During 2020–21 the delinquency-status codes additionally reflect COVID-era forbearance, which manifests as unusual persistence in delinquency states followed by elevated cure rates; no special treatment is applied, but results involving those years are read with this in mind.

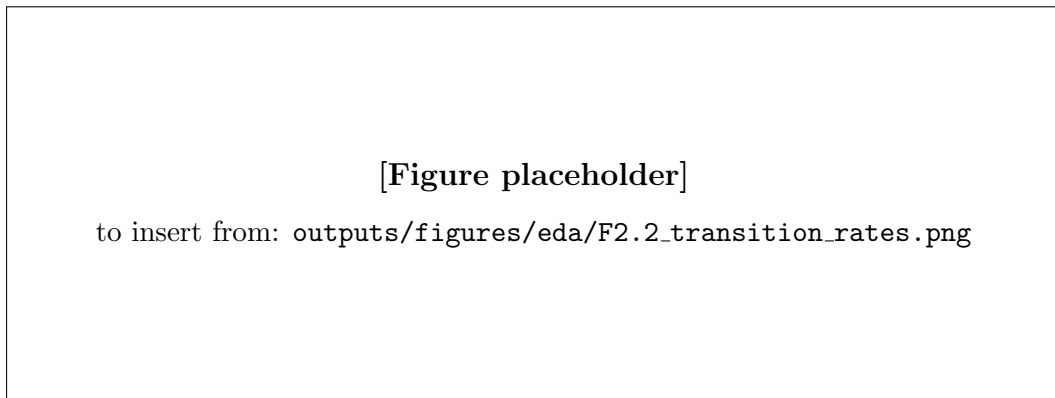


Figure 3.1: Monthly transition rates out of the Current state, 2000–2025: prepayment and entry into 30 DPD (left axis), serious-delinquency terminations (right axis). Vertical shading marks the 2008–11 crisis and the 2020–21 forbearance episode.

[Table placeholder]

to insert from: `outputs/tables/eda/T1.2_covariate_summary.tex`

Table 3.5: Summary statistics of model covariates (mean, standard deviation, 1st/50th/99th percentiles, missingness rate) over the full panel.

3.6 Exploratory Evidence: Nonlinearity and Interactions

Before any model is fitted, the case that nonlinear modelling is *needed* can be made by counting alone. The instrument is the empirical hazard curve: bucket a covariate into equal-population bins and plot the realised one-month transition rate per bin, with Wilson 95% confidence intervals. Because each curve is a single out-of-core aggregation over the columnar panel, the entire programme runs in minutes.

3.6.1 Single-covariate nonlinearity

Three hazard curves carry the argument. *Prepayment versus loan age* (Figure 3.2): the rate rises from 0.27% per month at origination to a peak of 1.56% at approximately 17 months of age, then decays to 1.11% by month 52 — the classic “seasoning hump,” strongly non-monotone, with a secondary late-life rise among long-surviving loans. *Prepayment versus the refinancing incentive* (Figure 3.3): the rate is flat at 0.4–0.6% per month while the option is out of the money, turns up sharply through zero, peaks at 2.71% near +1.5 percentage points of incentive, and falls back to 2.16% by +2.7 — a roughly sevenfold range containing both an inflection and a turning point, reproducing the flagship nonlinearity of Sirignano et al. [5] on this cleaner credit box. *Delinquency versus credit score* (Figure 3.4): the entry rate into 30 DPD falls twenty-four-fold, from 3.06% per month at FICO ≈ 623 to 0.13% at FICO ≈ 815 , and the decay is convex — the slope in the low-score half is roughly 3.7 times that in the high-score half. None of the three shapes is representable by a linear term.

3.6.2 Interactions: the case against additivity

A linear-*additive* model could in principle be patched with transformed single-variable terms; interactions are what it cannot absorb. On a credit-score $\times$ LTV grid (Fig-

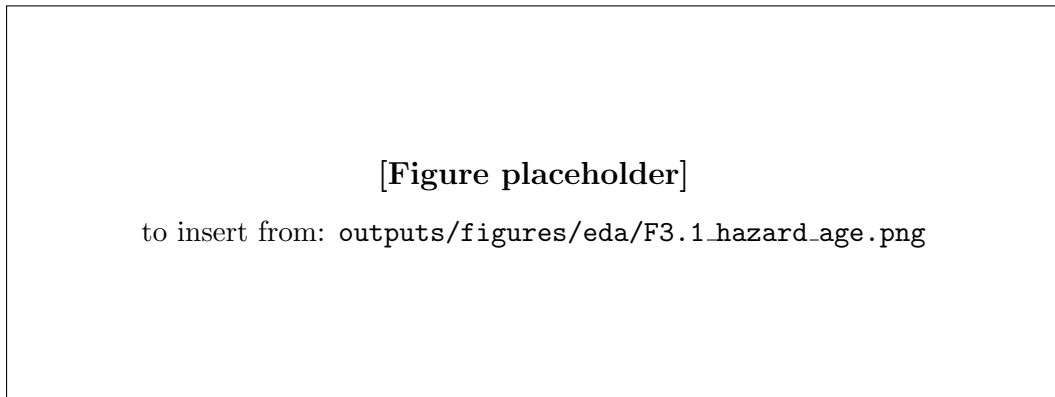


Figure 3.2: Empirical prepayment hazard by loan age (equal-population buckets, Wilson 95% intervals): the seasoning hump.

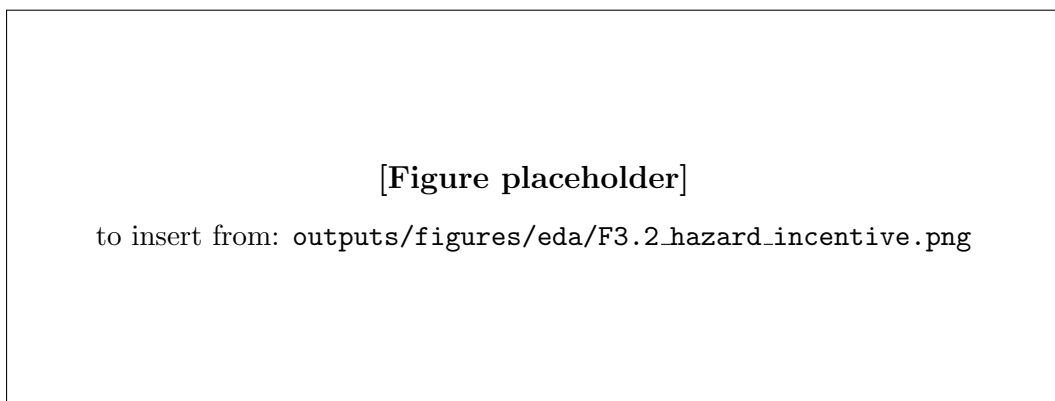


Figure 3.3: Empirical prepayment hazard by refinancing incentive: the S-curve, with burnout beyond +1.5 pp.

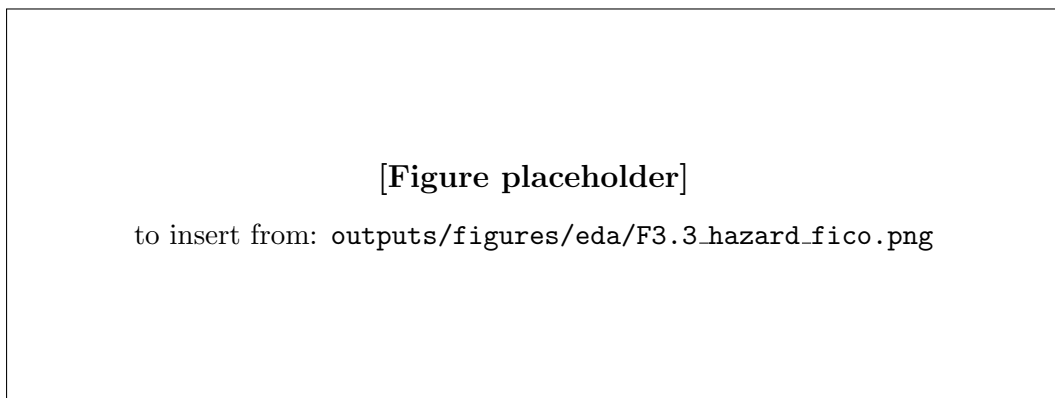


Figure 3.4: Empirical 30 DPD entry hazard by origination credit score: convex decay.

ure 3.5), the monthly rate of entry into delinquency ranges from 0.11% (best scores, lowest leverage) to 2.69% (worst scores, highest leverage) — and decisively, the *additive* credit-score gradient (worst-minus-best rate) grows from 1.73 percentage points at low LTV to 2.48 at high LTV, a 43% increase; equivalently, the leverage effect is roughly 8.5 times larger for low-score than high-score borrowers. Additivity forces this gradient to be constant; the violation sits exactly in the low-score/high-leverage corner that drives credit losses. In the prepayment channel (Figure 3.6), splitting the incentive S-curve by credit-score tercile shows the in-the-money refinancing response *rising* with credit quality — 2.36% per month in the lowest tercile against 3.01% in the highest — because credit-constrained borrowers cannot exercise the option: the incentive *slope* itself depends on the score. Finally, holding loan age fixed at 12–36 months to strip out seasoning, the delinquency hazard by origination cohort isolates a clean vintage effect: the 2007 cohort peaks at 1.68% per month against 0.47% for calm cohorts — a 3.6-fold premium (Figure 3.7) — so the origination vintage is a feature in its own right, not reducible to contemporaneous covariates.

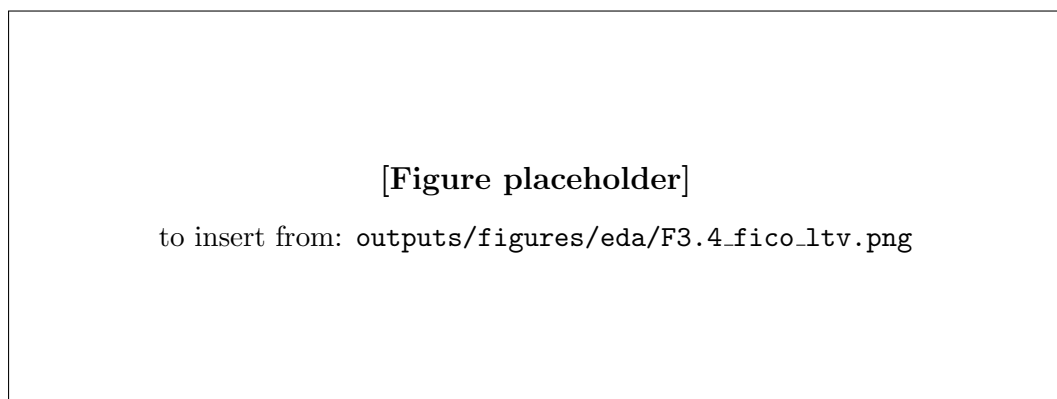


Figure 3.5: Monthly delinquency-entry rate on a credit-score $\times$ LTV grid (equal-population bands): the additive credit-score gradient grows with leverage — additivity is misspecified.

Taken together, the evidence points one way: monthly transition risk depends on the covariates nonlinearly (the seasoning hump, the incentive S-curve, convex credit-score decay) and interactively (score $\times$ leverage, incentive $\times$ score, vintage). A linear-additive model should therefore underfit — most severely on prepayment, the channel combining the sharpest single-variable nonlinearity with a first-order interaction — and Chapter 4 tests precisely this.

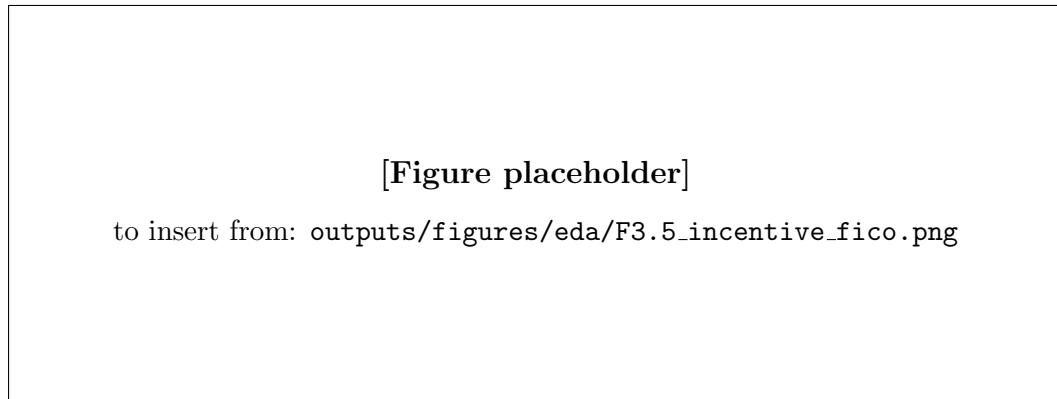


Figure 3.6: The incentive S-curve by credit-score tercile: the refinancing response to an in-the-money option rises with credit quality.

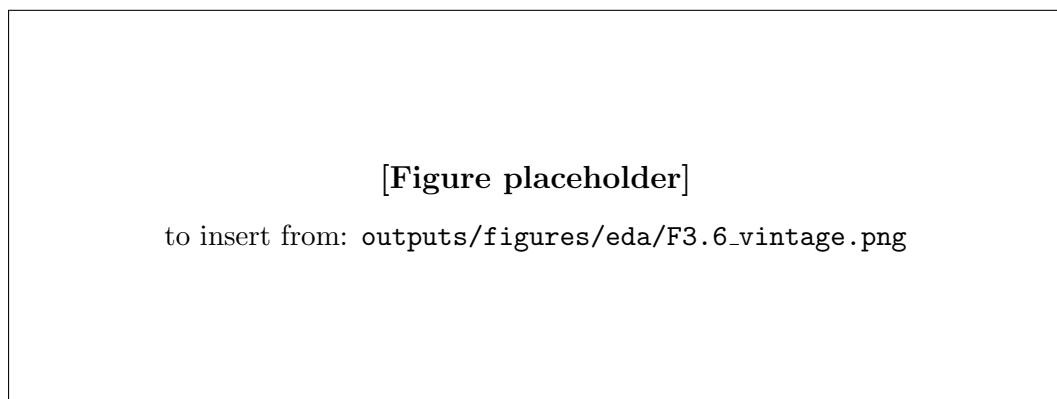


Figure 3.7: Delinquency hazard by origination cohort at fixed loan age (12–36 months): the 2006–08 bubble vintages.

3.7 Comparison to Sirignano et al. (2021)

The dataset employed in the present study differs in several important respects from the proprietary CoreLogic dataset used by Sirignano et al. [5], and these differences bear on the interpretation of the results.

Sirignano et al. [5] draw on records for approximately 120 million mortgages — both prime and subprime — originated across the United States between January 1995 and June 2014, yielding approximately 3.5 billion loan-month observations. The CoreLogic data encompass the full spectrum of mortgage products, including fixed-rate, adjustable-rate, hybrid, balloon, and interest-only loans, as well as government-insured instruments. Macro-economic covariates — zip-code-level house price indices from Zillow, county-level unemployment rates from the Bureau of Labor Statistics, and the Freddie Mac PMMS series — are merged at the loan-month level within the CoreLogic database. The dataset is proprietary and cannot be reproduced by independent researchers.

The Fannie Mae primary dataset, by contrast, is limited to conventional fixed-rate fully-amortising loans and excludes both subprime and government-insured products. Macro-economic covariates must be sourced and joined externally, as described in Section 3.3.3. These differences have two principal consequences. First, the estimation sample is more homogeneous in product mix; the nonlinear prepayment dynamics associated with adjustable-rate burnout and subprime option exercise — prominent in Sirignano et al. — are absent. Second, the absence of a subprime segment limits the range of credit-risk outcomes observed; Sirignano et al. [5] report a median FICO score of 630 for their subprime sub-sample against 730 for the prime sub-sample, whereas the Fannie Mae population is concentrated at the higher end of this credit-quality spectrum.

These limitations notwithstanding, the Fannie Mae dataset provides a fully reproducible basis for the analysis. The core methodological contribution of Sirignano et al. [5] — applying deep learning to a multinomial competing-risks model of monthly loan-state transitions — transfers completely to the Fannie Mae panel. The architecture, estimation procedure, and evaluation framework described in subsequent chapters are direct adaptations of that approach. Table 3.6 summarises the key differences.

Dimension	Sirignano et al. (2021)	Present study
Data source	CoreLogic (proprietary)	Fannie Mae SFLP (public)
Total loans	~120 million	57.6 million
Loan-month observations	~3.5 billion	3.31 billion
Origination window	1995–2014	1999–2025 (full vintage set, reporting to 2025-12)
Products covered	Fixed, ARM, hybrid, balloon, IO	Fixed-rate, fully amortising only
Credit segments	Prime and subprime	Conventional (predominantly prime)
Macro covariates	Embedded (Zillow, BLS, PMMS)	External join required (FHFA HPI, PMMS)
Geographic detail	Full 5-digit zip code	3-digit truncated zip; MSA
States modelled	Seven	Seven (identical mapping)
Reproducibility	Not publicly reproducible	Fully reproducible

Table 3.6: *Comparison of the Fannie Mae SFLP primary dataset and the CoreLogic dataset used by Sirignano et al. [5].* CoreLogic counts are approximate as reported by the authors; Fannie Mae counts are measured on the assembled panel (Section 3.5).

Chapter 4

Results

All results in this chapter are out of sample under the rolling design of Section 2.4 unless explicitly labelled in-sample. Negative log-likelihood (NLL) is reported as the mean of (2.8) over the relevant frozen test rows; lower is better.

4.1 Architecture selection and benchmarks on the tuning window

Hyperparameter and architecture selection was carried out once, on the $k=2015$ tuning window (training on labels through December 2013, early stopping and selection on validation year 2014), and frozen thereafter. The search covered depth $L \in \{1, 3, 5, 7\} \times \text{dropout} \{0, 0.2, 0.5\}$ at zero weight decay, plus an ℓ_2 sweep at the leading configuration — a pruned grid of 14 cells anchored at the five-layer geometry of Sirignano et al. [5].

[Table placeholder]

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Table 4.1: Tuning-window comparison (the analogue of Table 11 in 5): in- and out-of-sample NLL for the empirical matrix, bucketed matrix, multinomial logit, augmented logit, neural networks of depth 1/3/5/7, the selected configuration with dropout, and the eight-network ensemble.

Three findings from the grid, at dev scale (a stratified ~ 3.15 -million-row training slice of the window):

- **Regularisation moves the optimum deeper.** Without dropout the validation-optimal depth is one hidden layer; with dropout 0.2 it shifts to three — qualitatively replicating the dropout–depth interaction documented by Sirignano et al. [5]. A small ℓ_2 penalty (10^{-5}) partially rescues the five-layer network (validation NLL $0.09767 \rightarrow 0.09630$) but does not overturn the ordering.
- **The selected configuration** is depth 3, dropout 0.2: validation NLL 0.09602, test NLL 0.10319 on the tuning window. In-sample NLL exceeding out-of-sample in parts of the grid is expected here, not inverted overfitting: the expanding training window (2000–2013) contains the crisis years while the 2015 test year is calm.
- **The ensemble helps, with diminishing returns.** The eight-network ensemble at the selected configuration reaches test NLL 0.10274, beating both the average single member (0.10359) and the best individual member (0.10307); the ensemble-size curve flattens by roughly five members (Figure 4.1).

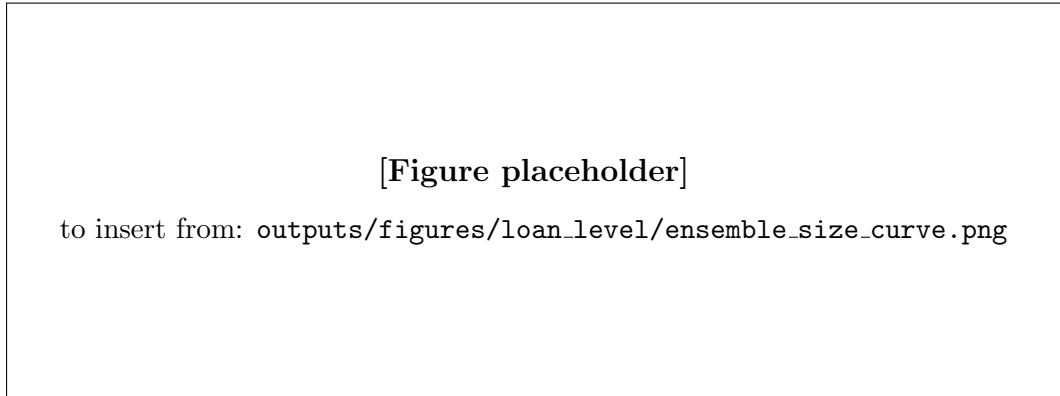


Figure 4.1: Out-of-sample NLL versus number of ensemble members, tuning window (the analogue of Figure 7 in 5).

The multinomial logit baseline was validated against a standard library implementation (agreement in NLL to within 10^{-3} across the regularisation grid; the loop’s embedding-based reimplementation subsequently matched it to within 10^{-4}), so the gaps between logit and network rows in Table 4.1 are attributable to architecture alone.

Full-scale depth check. Because the grid above was run at dev scale, the depth choice was re-examined at full scale ($\sim 10^8$ training rows) before the rolling loop was launched: depth 3 and depth 5 were re-fitted on the full tuning window (with and

without the ℓ_2 rescue). Depth 3 won robustly, and the frozen configuration for the loop is depth 3, dropout 0.2, weight decay 10^{-5} . This is a substantive deviation from Sirignano et al. [5], whose optimum at population scale was five layers: on the prime conforming credit box, the optimal network is *shallower* — the nonlinearity premium documented in Section 3.6 is real but structurally simpler than in subprime-heavy private-label data.

4.2 The rolling backtest, 2015–2025

[Pending] This section is populated by the rolling loop and evaluation suite (tasks M10b–M11), which at the time of writing are running. Structure below; numbers to follow.

The headline exhibit is the test-year $\times$ model NLL matrix — empirical matrix, multinomial logit, the frozen network, and (on the five key windows) the ensemble, for each test year 2015 through 2025 — together with the pooled all-years NLL, in which every test row is predicted exactly once by a model that never saw it.

[Table placeholder] to insert from: <code>outputs/tables/loan_level/table.b.tex</code>

Table 4.2: Out-of-sample NLL by test year and model, 2015–2025, with the pooled all-years comparison; by-origin-state breakdown beneath.

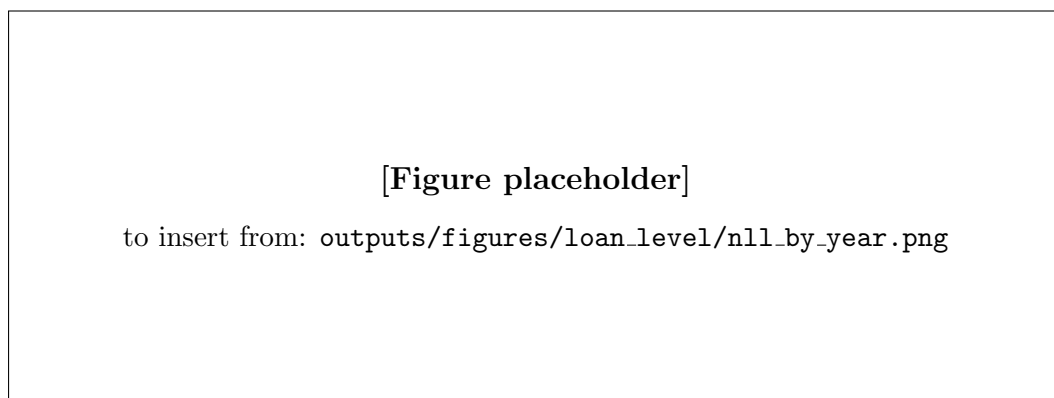


Figure 4.2: Out-of-sample NLL by test year, one line per model: the time series of the value of nonlinearity.

[Table placeholder] to insert from: <code>outputs/tables/loan_level/auc_table.tex</code>

Table 4.3: One-versus-rest AUC by origin and destination state, pooled test predictions: logit versus network versus ensemble.

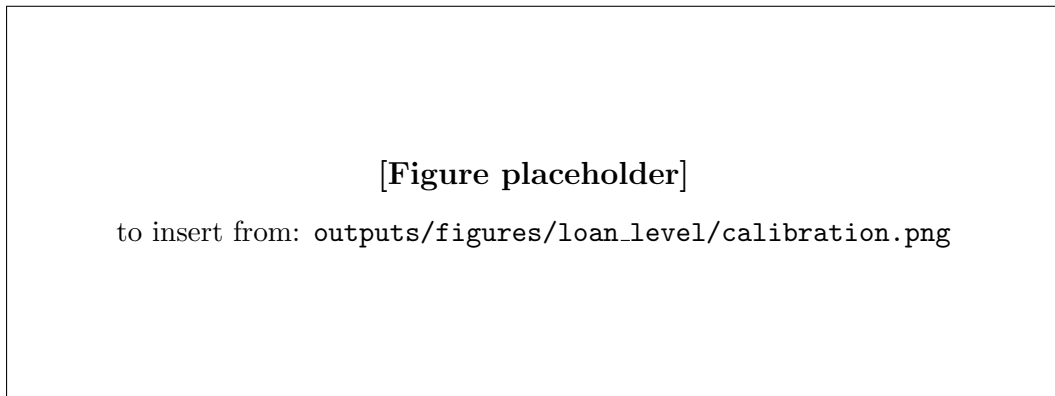


Figure 4.3: Reliability diagrams for the Current→Prepaid and Current→30 DPD transitions, pooled and for the 2020–21 forbearance years separately; predicted versus realised aggregate transition rates over 2015–2025.

Robustness exhibits (task M12): seed variance of the selected network against the network–logit gap; stability of the model ranking across the eleven test years (notably through 2020–21 and 2022–23); re-tuning sensitivity on the $k=2019$ window; and the width sweep at the selected configuration (Appendix A).

4.3 Pool-level results and economic translation

[Pending] This section is populated by Phase 3 (tasks M13–M15) once the rolling backtest completes. Structure below.

Pool-level accuracy: predicted versus realised twelve-month prepayment and serious-delinquency counts for characteristic-bucket and random pools at the window anchors, per model (Figure 4.4; Table 4.4), with interval coverage against (2.10) and the by-anchor accuracy series as the regime exhibit.

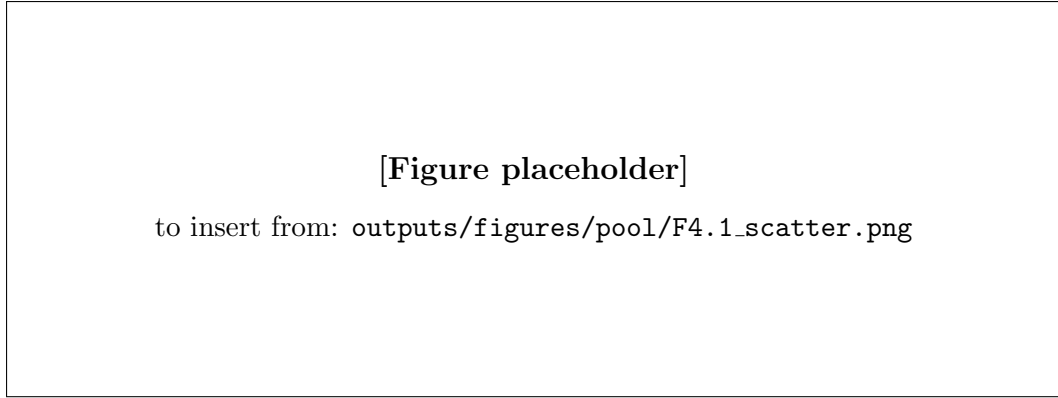


Figure 4.4: Predicted versus realised twelve-month prepayment counts, random pools, by model.

[Table placeholder]

to insert from: `outputs/tables/pool/T4.2.tex`

Table 4.4: Pool-level R^2 and RMSE by model, outcome, and anchor; interval coverage.

Economic translation (Section 2.7.3): speed, timing, and price errors per pool $\times$ model $\times$ anchor, with the headline being the ensemble-versus-logit reduction in mean absolute price error, and the bucket-level price-error map localising the linear model’s mispricing.

[Table placeholder]

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Table 4.5: CPR, weighted-average-life, and price errors by model and anchor.

[Figure placeholder]

to insert from: `outputs/figures/pool/F5.2_price_error_buckets.png`

Figure 4.5: Price error by characteristic bucket and model: where the linear model misprices.

Chapter 5

Conclusions

Appendix A

Supplementary tables and figures

Appendix B

Reproducibility

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